Victoria Knapp Perez

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Summary

Physics Ph.D. candidate with a strong background in machine learning, reinforcement learning, and data-driven modeling, with research experience in theoretical particle physics. Experienced in AI methods such as reinforcement learning (OpenAI Gym), deep learning (PyTorch), and high-performance computing, with an emphasis on automated scientific model discovery and symbolic reasoning. I aim to apply my AI expertise and problem-solving capabilities to impactful, real-world challenges.

Education

- **University of California, Irvine, USA,** Ph.D. Physics (GPA=4.00) **(Expected) June 2026**
- **University of California, Irvine, USA,** M.S. Physics (GPA=4.00) **2023**
- **National Autonomous University of Mexico, Mexico,** B.Sc. Physics (GPA=9.88/10), Cum Laude **2020**

Skills

- Programming & Platforms: Python (7+ years), Java (1+ year), Bash scripting (5+ years), LaTeX (9+ years), Slurm (2+ years), CUDA, and High-Performance Computing (HPC) environments.
- Frameworks and Libraries: PyTorch, FastAI, Stable Baselines, Hugging Face Transformers, scikit-learn, OpenAI Gym, Pandas, NumPy, Matplotlib.
- Machine Learning & Data Science: Reinforcement learning (PPO), deep learning (CNNs, RNNs, transformers, GNNs), attention and sequence-to-sequence architectures, and fine-tuning of pretrained models such as BERT and GPT.
- Development Tools & Version Control: Git, Jupyter Notebook, Linux shell scripting.
- Certifications: Coursera Java (2020), Deep Leaning(2025).

Work Experience

University of California: Irvine, USA **2021-Present**

Research Assistant & Teaching Assistant

- Engineered end-to-end ML and RL pipelines to automate physics-model construction, integrating PPO agents (Stable Baselines) with HPC training environments (Python, Mathematica, Slurm). Reduced evaluation runtime and enhanced model performance through parallelized optimization.
- Published peer-reviewed papers on theoretical particle physics, cosmology and integrating ML methods into physics model discovery. Collaborated with cross-disciplinary teams on ML pipeline design.
- Delivered instruction and mentorship across 10+ courses from high-school to PhD level, developing strong communication and leadership skills while simplifying complex topics for diverse audiences.

National Autonomous University of Mexico: Mexico City, Mexico **2020-2021**

Research Assistant & Teaching Assistant

- Built Python and Mathematica workflows to simulate and analyze physical models, producing efficient automation for large numerical experiments.
- Delivered instruction across 5 undergraduate level courses.

Selected projects

AI-assisted Neutrino Flavor Theory Design **-** University of California: Irvine **2025**

- Developed the "Autonomous Model Builder (AMBer)" framework, a reinforcement-learning system built with Stable Baselines and integrated into a custom end-to-end software pipeline to automate neutrino physics-model generation, parameter optimization, and validation against experimental data. Enabled large-scale exploration of theoretical spaces and reduced manual analysis time.
- Created and maintained FlavorBuilder, a Python package supporting model construction, parameter optimization, and data-driven validation.
- Tech stack: Python, PyTorch, Stable Baselines, OpenAI Gym, reinforcement learning (PPO), data analysis, high-performance computing (Slurm).

Text Reliability Classification with Fine-Tuned LLMs - Erdos Institute ML Project **2025**

- Fine-tuned a LSTM model for multi-class truthfulness prediction on Kaggle news datasets, benchmarking against a TF-IDF + logistic regression baseline to quantify LSTM advantages.
- Tech stack: Python, PyTorch, Hugging Face Transformers, SpaCy, scikit-learn, Pandas, NumPy.

Selected publications and awards

- J. B. Baretz, M. Fieg, V. Ganesh, A. Ghosh, V. Knapp-Perez, J. Rudolph and D. Whiteson, "Towards AI-assisted Neutrino Flavor Theory Design", [arXiv:2506.08080 [hep-ph]]. **2025**
- Latino Excellence and Achievement Award (LEAD), honored for research excellence and leadership of Hispanic/Latinx students at UCI. **2024**
- UC President's Lindau Nobel Laureate Meetings Fellowship, selected to engage with Nobel Laureates and international scientists. **2024**
- UCI Department of Physics and Astronomy Outstanding Graduate Student Award, recognizing top first-year graduate student. **2022**

Physics Publications

1. **V. Knapp-Perez**, G. Mohlabeng, M. Ratz and T. M. P. Tait, “Thermal History of Non-equilibrated Scalars,” [arXiv:2507.21523 [hep-ph]].
2. F. Huang and **V. Knapp-Perez**, “Cosmological stasis from field-dependent decay,” JHEP 09, 072 (2025) doi:10.1007/JHEP09(2025)072 [arXiv:2502.20449 [hep-ph]]
3. A. Baur, M. C. Chen, **V. Knapp-Perez** and S. Ramos-Sanchez, “Modular flavored dark matter,” JHEP 12, 091 (2024) doi:10.1007/JHEP12(2024)091 [arXiv:2409.02178 [hep-ph]].
4. **V. Knapp-Perez**, X. G. Liu, H. P. Nilles, S. Ramos-Sanchez and M. Ratz, “Matter matters in moduli fixing and modular flavor symmetries,” Phys. Lett. B 844, 138106 (2023) doi:10.1016/j.physle tb.2023.138106 [arXiv:2304.14437 [hep-th]].
5. . M. C. Chen, **V. Knapp-Perez**, M. Ramos-Hamud, S. Ramos-Sanchez, M. Ratz and S. Shukla, “Quasi–eclectic modular flavor symmetries,” Phys. Lett. B 824, 136843 (2022) doi:10.1016/j.physletb .2021.136843 [arXiv:2108.02240 [hep-ph]].
6. Y. Almumin, M. C. Chen, **V. Knapp-Pérez**, S. Ramos-Sánchez, M. Ratz and S. Shukla, “Metaplectic Flavor Symmetries from Magnetized Tori,” JHEP 05 (2021), 078 doi:10.1007/JHEP05(2021)078 [arXiv:2102.11286 [hep-th]].
7. A. Ayala, S. Hernandez-Ortiz, L. A. Hernandez, **V. Knapp-Perez** and R. Zamora, “Fluctuating temperature and baryon chemical potential in heavy-ion collisions and the position of the critical endpoint in the effective QCD phase diagram,” Phys. Rev. D 101 (2020) no.7, 074023 doi: 10.1103/PhysRevD.101.074023 [arXiv:2003.04505 [hep-ph]].